

# Learning with AI

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*Opportunities, limits, and open questions from AI-assisted research workflows*

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# Two genuine strengths

## Information retrieval

- Like a far more powerful search engine
- Finds, summarizes, and connects sources fast
- Surfaces work you would never have found

## Coding

- Like a strong, tireless programmer
- Writes, refactors, and debugs quickly
- Deploys an analysis in minutes, not days

On these two, AI is genuinely strong. Start there.

# But a strength is not the whole job

## Retrieval ≠ thinking

- Finding sources is not logical inference
- It is not creating a new argument
- Retrieval gives you inputs — not conclusions

## Coding ≠ design

- Writing code is not empirical design
- The hard part is what to estimate, and why
- Code executes a design; it doesn't choose one

The strengths sit one level below the real work.

# Two hurdles you cannot outsource

## Make sense of the theory

- Different fields, different vocabularies for one problem
- Can you hold them together in your own mind?
- Integration is judgment — not retrieval

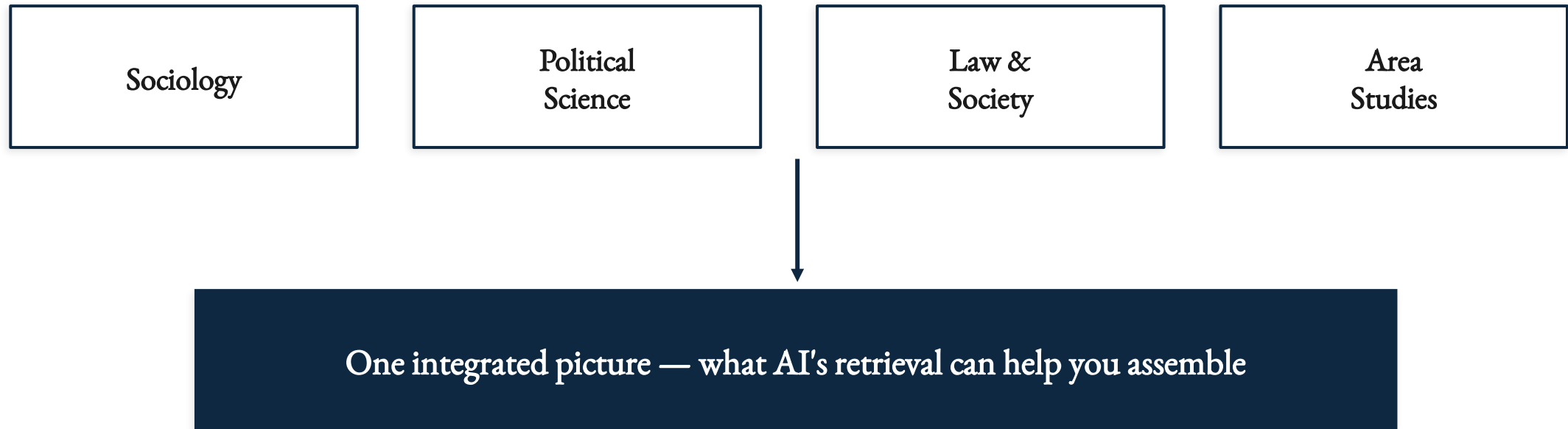
## Comprehend the code

- Can you read the stack the agent built?
- Can you evaluate what it actually does?
- Can you defend every modeling choice in it?

AI raises the ceiling — only if you can reach it.

# Organizing interdisciplinary literature

*Many research topics are balkanized — one question, scattered across fields.*



**AI can cross silos faster than any of us — if you can make sense of them.**

# Handle the empirical challenge properly

**Even before AI:** coding 2SLS (two-stage least squares) in R was easy. Deciding how to handle the endogeneity was the hard part.

**Now, with AI:** it can pick the model and write the code. Do you know what your results mean?

You can outsource coding, writing, even part of thinking — but not the deeper work of understanding.

# “It can — but suboptimally”

Can AI agents do the harder tasks — inference, design — themselves?

~93%

of the way there — close to the conventional standard, but not at it. And maybe never.

*(my argument — a working estimate, not a measured benchmark)*

**Close enough to rely on. Not close enough to stop thinking.**

# Two moves that pay off

## Retrieval → reach

- Expand your theoretical landscape
- Pull scholarship across fields together
- See the whole conversation, not your silo

## Coding → speed

- Quickly deploy the design you have
- Evaluate the potential issues
- Report the findings back, fast

Use AI to widen what you reach and quicken what you test.



# Practice 1 — ask the right questions

## For human researchers

- Understand the theoretical context
- Understand the empirical design
- The question is the research

## For AI

- ...the same understanding makes a better prompt
- Good question in → good work out
- Vague ask in → confident nonsense out

**A good prompt is just a well-posed research question.**

## Practice 2 — read more, not less

Since I started working with AI (2021), I have read **more** theory — not less.

### Why the counterintuitive direction?

- To ask better questions, I need deeper context
- To judge what AI surfaces, I need my own grounding
- To integrate across fields, I have to actually know them

**AI handles the mechanical — which frees time for the conceptual.**

# Practice 3 — learn the unknowns AI surfaces

*When the agent suggests something I don't know, that is the assignment — go learn it.*

## New data

a source you'd never have reached

## New model

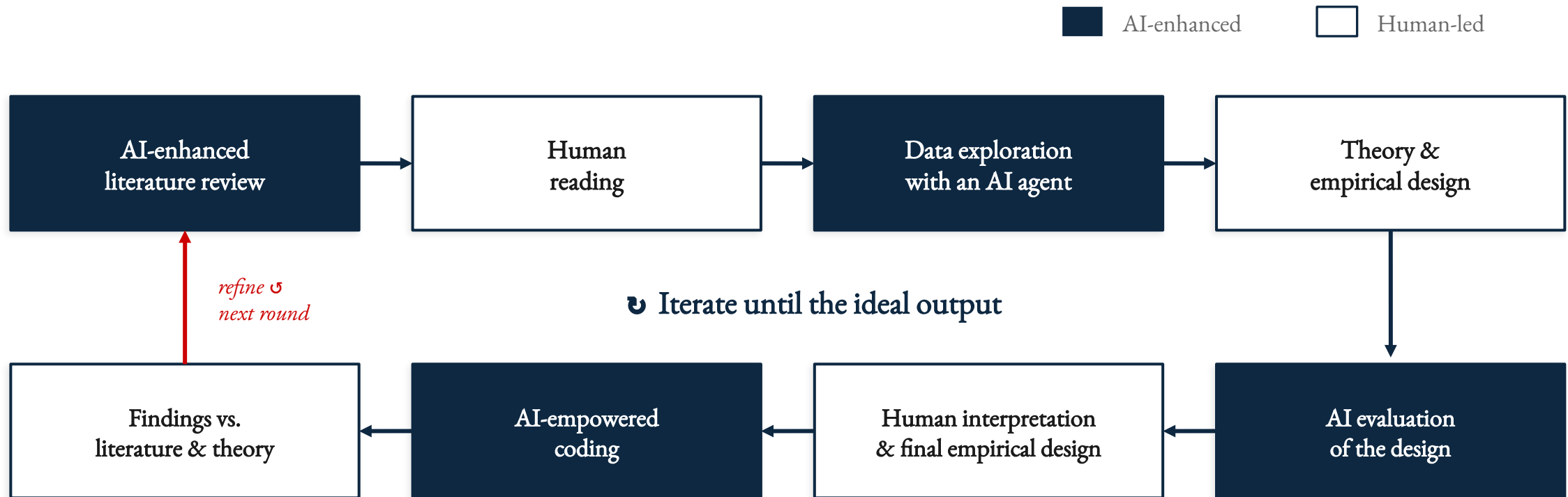
a specification you didn't know fit

## New test

a robustness check worth running

**Treat every unfamiliar suggestion as something to learn — not just to accept.**

# An AI-enhanced research loop



*Human-led steps — reading, design, interpretation, the final draft — anchor every loop; AI accelerates the rest.*

**AI moves through every step — but you own the reading, the design, and the meaning.**

# The ideal output package (Liu et al. 2026)

*I agree with their direction: redesign the artifact so nothing checkable is lost.*

## Two hidden taxes today

- Storytelling tax — failed experiments vanish to fit a linear narrative
- Engineering tax — prose drops the technical detail agents need

## The fix — agent-native research artifacts:

Scientific logic

Fully-specified executable code

Exploration graph — keeps the dead ends & failures

Raw evidence behind every claim

*Built for checking — agents answer questions about the paper far better (72.4% → 93.7%).*

**Keep the failures and the full detail — a paper built to be checked, by humans and agents.**

# My robustness battery (CopChopper)

*What “keep every check” looks like in one of my own projects:*

```
copchopperGIS/results/
├── tables/comprehensive/    – robustness batteries
│   ├── AppA_robust_strict    strict-window refit
│   ├── AppB_robust_loiter    loiter outcome
│   ├── AppC_robust_cell_eventmonth cell × event-month FE
│   ├── AppD_robust_h4_alternatives alternative H4 specs
│   ├── AppE_robust_misc      misc. sensitivity
│   └── AppF_h1_permutation    permutation inference
├── tables/individual/      – appendix battery (A1-A29):
│   ├── Conley spatial SE · spatial spillover · PPML per agency ·
│   ├── grid PPML · alt-grid · day-of-week & holiday FE · SMD balance ·
│   └── fleet roster & capacity · subsample · admin asymmetry · ...
└── figures/                – event-study leads/lags · placebo-battery
    ├── null distributions · family/admin forest plots · spec panels
```

**Liu’s ideal package, in miniature: keep every check — and ship it.**

# What means empirically rigid?

## Reproducibility

- “The same-same test”
- Same data, same code
- Same result?

## Robustness

- “The what-if test”
- Same data, other methods
- Conclusion survives?

## Replicability

- “The new-data test”
- New data, same question
- Finding holds?

*Most published work holds up — but checking it is slow human work AI can now carry.*

**These tests are mechanical by design — exactly where AI helps.**

# What good AI-era research looks like

*Two pillars — the understanding, and the proof:*

## Solid interdisciplinary vision & framing

- Cross-field theory, integrated — not siloed
- The understanding AI cannot supply
- This is what you bring to the table

## A valid supporting package

- Reproducible, replicable, robust
- Built so others — and agents — can check
- The agent-native artifact we just saw

**Framing carries the understanding; the package carries the proof.**



# Open questions — for this community

- ? Where does the durable ~93% gap actually sit — and does it move?
- ? How do we train comprehension when output is cheap?
- ? What should count as a valid supporting package now?
- ? Where does authorship — and responsibility — sit in an AI-assisted paper?

# Thank you

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## References

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*Slides built with — and audited by — the multi-agent workflow described in this talk.*

# Opportunities · Limits · Questions

BACKUP

## Opportunities

- Cross-silo theoretical reach
- Faster deploy–evaluate–report
- Verification carried at scale

## Limits

- Retrieval  $\neq$  inference
- Coding  $\neq$  design
- A durable ~93% gap

## Questions

- What is a valid package now?
- Where does authorship sit?
- How do we train comprehension?